Status	Finished
Started	Sunday, 18 October 2026, 3:48 PM
Completed	Sunday, 18 October 2026, 4:03 PM
Duration	15 mins 14 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Complete

Mark 1.00 out of 1.00

How does Merge Sort achieve its efficiency?

- a. By breaking the input into smaller parts and merging them
- b. By sorting data in a single pass
- c. By comparing elements sequentially
- d. By using the bubble-up method

The correct answer is: By breaking the input into smaller parts and merging them

Question **2**

Complete

Mark 1.00 out of 1.00

Which built-in Python function is used to sort data?

- a. sort()
- b. order()
- c. sorted()
- d. arrange()

The correct answer is: sorted()

Question **3**

Complete

Mark 1.00 out of 1.00

What does the Bubble Sort algorithm primarily focus on during each pass?

a. Dividing the list into halves
b. Bubbling up the smallest element
c. Sorting the entire list in one pass
d. Bubbling up the largest element to its correct position

The correct answer is: Bubbling up the largest element to its correct position

Question **4**

Complete

Mark 1.00 out of 1.00

What is mean by stable sorting algorithm?

a. A sorting algorithm is stable if it doesn't preserver the order of duplicate keys
b. A sorting algorithm is stable if it preserves the order of non-duplicate keys
c. A sorting algorithm is stable if it preserves the order of all keys
d. A sorting algorithm is stable if it preserves the order of duplicate keys

The correct answer is:
A sorting algorithm is stable if it preserves the order of duplicate keys

Question **5**

Complete

Mark 1.00 out of 1.00

Which algorithm is efficient for analyzing the frequency distribution of items in a list?

a. Quick Sort
b. Bubble Sort
c. Merge Sort
d. Linear Search

The correct answer is: Merge Sort

Question **6**

Complete

Mark 1.00 out of 1.00

What is sorting in the context of computer science?

- a. Inserting data into a list
- b. Searching for data in a list
- c. Deleting data from a list
- d. Arranging data in a particular format

The correct answer is: Arranging data in a particular format

Question **7**

Complete

Mark 1.00 out of 1.00

Which of the following is not an in-place sorting algorithm?

- a. Merge sort
- b. Quick sort
- c. Selection sort
- d. Heap sort

The correct answer is:
Merge sort

Question **8**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the process of Merge Sort?

- a. It repeatedly finds the minimum element and moves it to the sorted part of the list
- b. It divides the list into two halves, sorts each half, and then merges them
- c. It builds a sorted array one element at a time
- d. It compares adjacent elements and swaps them if necessary

The correct answer is: It divides the list into two halves, sorts each half, and then merges them

Question **9**

Complete

Mark 1.00 out of 1.00

What is one of the key advantages of using the built-in sorted() function in Python?

a. It requires external libraries
b. It is less efficient than custom sorting algorithms
c. It sorts data out of the box efficiently
d. It only works with integer arrays

The correct answer is: It sorts data out of the box efficiently

Question **10**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the term "sorting" in computer science?

a. Arranging data in a specific order
b. Finding a specific element in a list
c. Merging two datasets
d. Removing duplicates from a list

The correct answer is: Arranging data in a specific order

Question **11**

Complete

Mark 1.00 out of 1.00

_____is putting an element in the appropriate place in a sorted list yields a larger sorted order list.

a. Selection
b. Distribution
c. Extraction
d. Insertion

The correct answer is:
Insertion

Question **12**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is based on repeatedly dividing the list into halves?

a. Quick Sort
b. Insertion Sort
c. Merge Sort
d. Bubble Sort

The correct answer is: Merge Sort

Question **13**

Complete

Mark 1.00 out of 1.00

_____ explain how an algorithm will perform when the input grows larger.

a. Sorting
b. Complexity
c. Merging
d. Searching

The correct answer is:
Complexity

Question **14**

Complete

Mark 1.00 out of 1.00

What is the primary benefit of using sorting algorithms in programming?

a. Makes code execution slower
b. Makes data harder to manage
c. Provides a basis for other algorithms to work efficiently
d. Decreases the efficiency of algorithms

The correct answer is: Provides a basis for other algorithms to work efficiently

Question **15**

Complete

Mark 1.00 out of 1.00

Which of the following is a key reason for the importance of sorting algorithms?

a. Sorting makes it harder to search for items
b. Sorting is rarely used in programming
c. Sorting decreases the efficiency of selection operations
d. Sorting helps in finding duplicates quickly

The correct answer is: Sorting helps in finding duplicates quickly